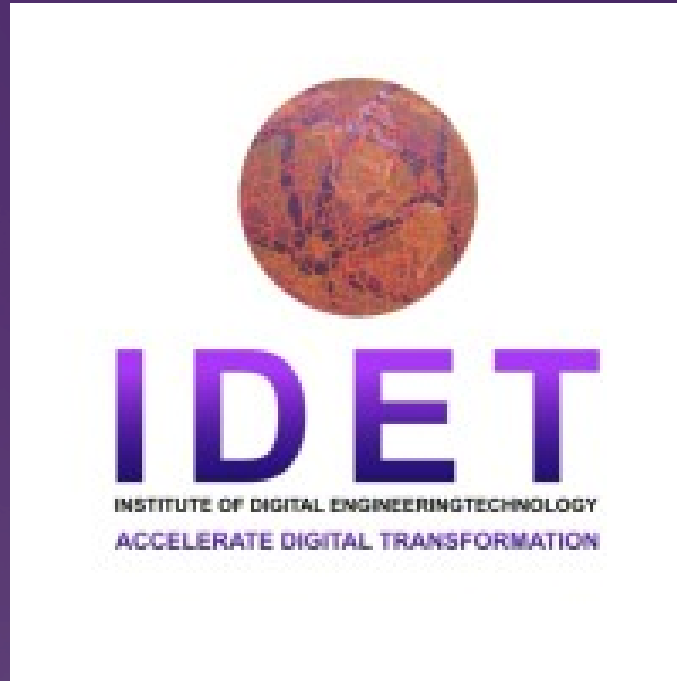




IDET Certified: Cloud Operations & AWS Practitioner Masterclass

Day 01

Theory

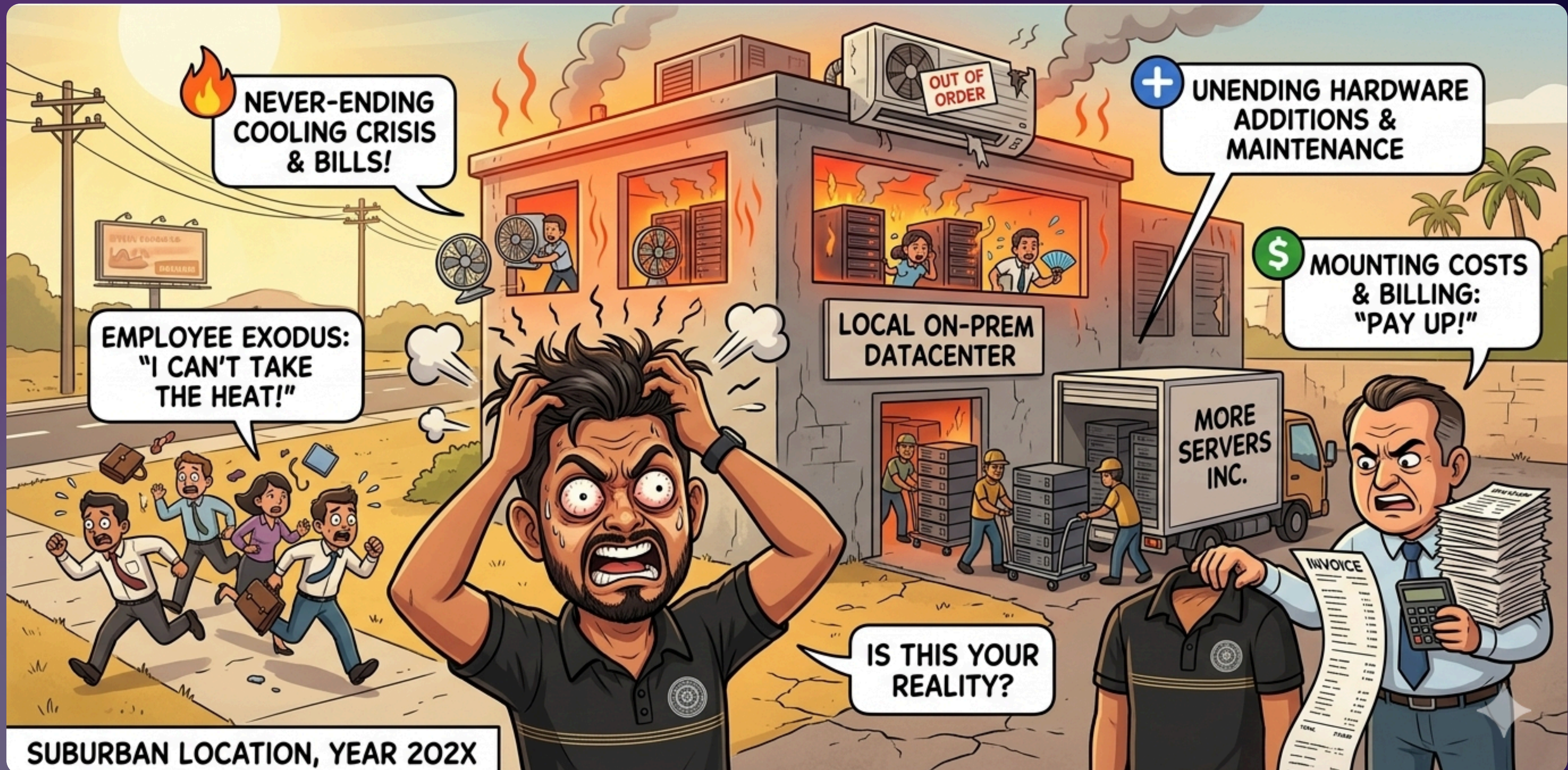
- What is Cloud Computing? Evolution from Traditional IT
- Service Models: IaaS, PaaS, SaaS with real-world examples
- Deployment Models: Public, Private, Hybrid, Multi-Cloud
- Benefits of Cloud Computing (Scalability, Cost, Flexibility)
- Introduction to AWS: History, Market Position, Global Infrastructure
- AWS Regions, Availability Zones, Edge Locations explained
- AWS Free Tier overview and account safety
- AWS Service Categories overview (Compute, Storage, Database, Networking, etc.)

Day 01

Practicals

- Lab 1.1: Create AWS Free Tier Account
- Lab 1.2: AWS Console Navigation & Service Discovery
- Lab 1.3: Set up billing alerts and budgets
- Lab 1.4: Explore AWS Global Infrastructure map
- Lab 1.5: AWS Pricing Calculator - Estimate costs for sample workloads

Traditional IT Infrastructure / Problems



Problems with traditional IT approach

- Pay for the rent for the data center
- Pay for power supply, cooling, maintenance
- Adding and replacing hardware takes time
- Scaling is limited
- Hire 24/7 team to monitor the infrastructure
- How to deal with disasters? (earthquake, power shutdown, fire...)

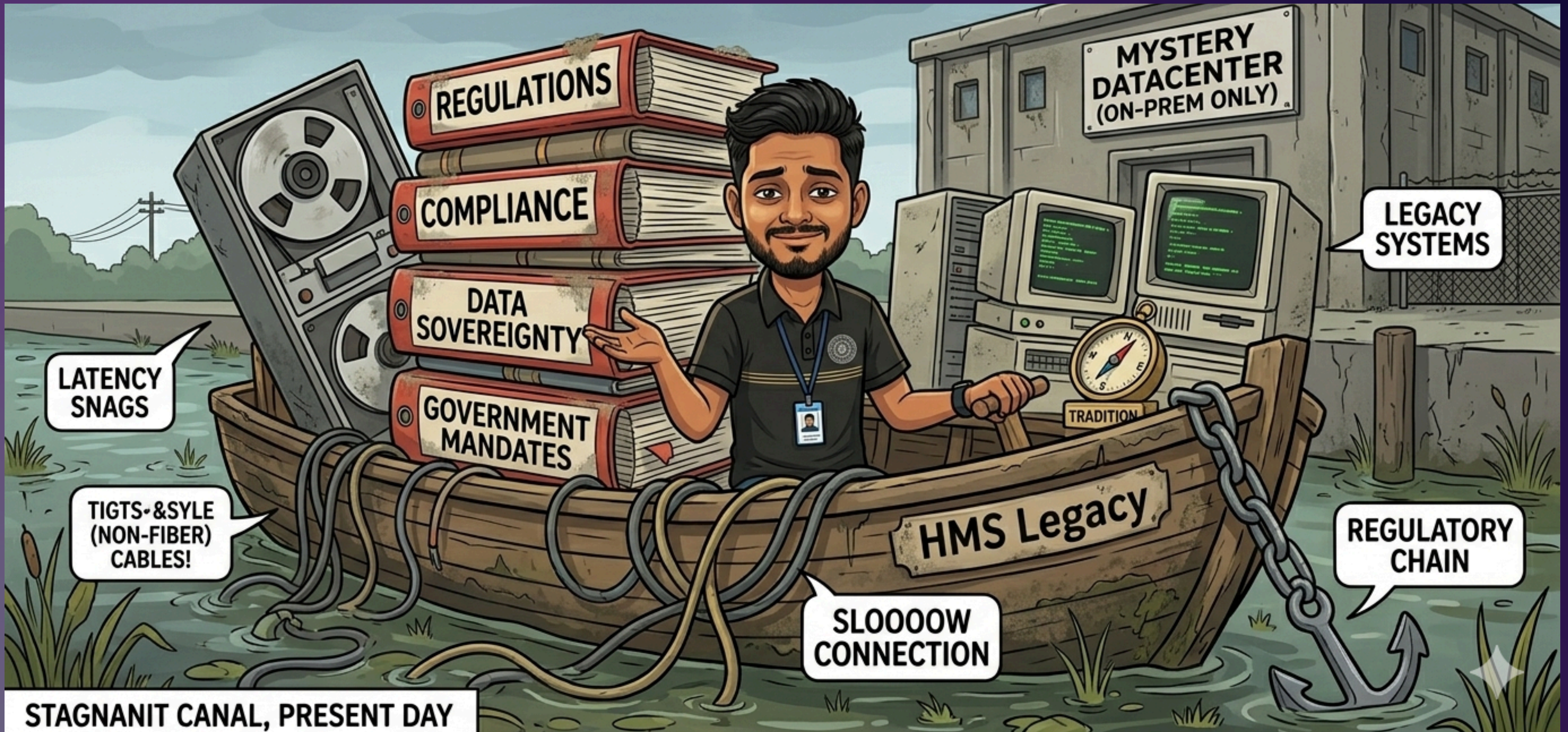
What is Cloud Computing?

- Cloud computing is the on-demand delivery of compute power, database storage, applications, and other IT resources
- Through a cloud services platform with pay-as-you-go pricing
- You can provision exactly the right type and size of computing resources you need
- You can access as many resources as you need, almost instantly
- Simple way to access servers, storage, databases and a set of application services

6 Advantages of Cloud Computing

- Trade capital expense (CAPEX) for operational expense (OPEX)
- Benefit from massive economies of scale
- Stop guessing capacity
- Increase speed and agility
- Stop spending money running and maintaining data centers
- Go global in minutes

Why Still On Prems?



The Deployment Models of the Cloud



The Deployment Models of the Cloud

Private Cloud

- Cloud services used by a single organization, not exposed to the public.
- Complete control
- Security for sensitive applications
- Meet specific business needs

Public Cloud

- Cloud resources owned and operated by a third party cloud service provider delivered over the Internet
- Six Advantages of Cloud Computing

Hybrid Cloud

- Keep some servers on premises and extend some capabilities to the Cloud
- Control over sensitive assets in your private infrastructure
- Flexibility and cost effectiveness of the public cloud

6 Advantages of Cloud Computing

- Trade capital expense (CAPEX) for operational expense (OPEX)
- Benefit from massive economies of scale
- Stop guessing capacity
- Increase speed and agility
- Stop spending money running and maintaining data centers
- Go global in minutes

Types of Cloud Computing

Infrastructure as a Service (IaaS)

Provide building blocks for cloud IT
Provides networking, computers, data storage space
Highest level of flexibility
Easy parallel with traditional on-premises IT

AWS EC2
Azure VM

Platform as a Service (PaaS)

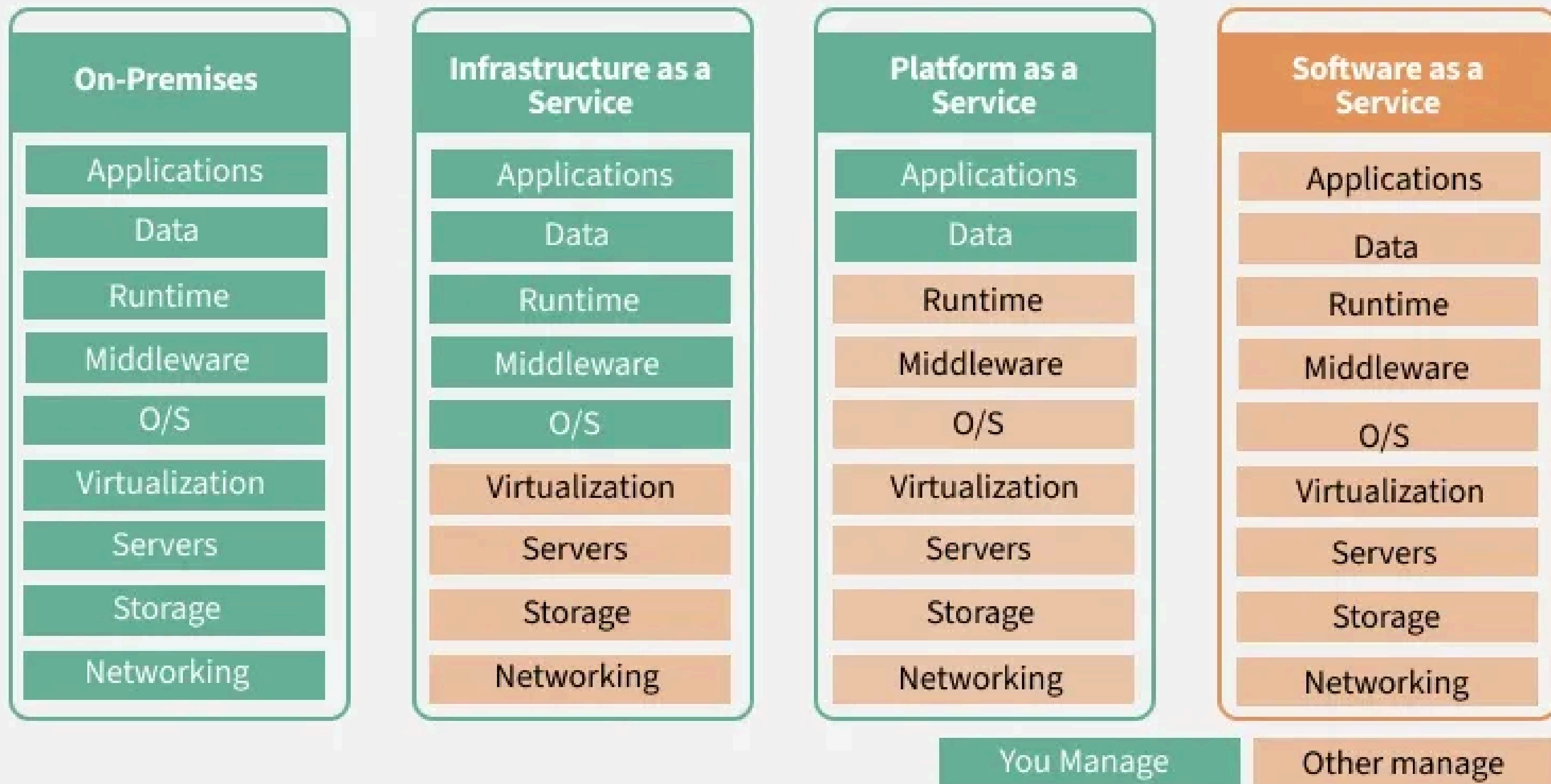
Removes the need for your organization to manage the underlying infrastructure
Focus on the deployment and management of your applications

Heroku
Google App Engine

Software as a Service (SaaS)

Completed product that is run and managed by the service provider

Google Docs,
Zoom, MS Teams



aws



Who is AWS?

- Amazon Web Services (AWS) is a leading cloud computing platform provided by Amazon.
- They Provide Mainly :

Compute

Storage

Databases

Networking

AI/ML, DevOps, Security

AWS Global Infrastructure

<https://aws.amazon.com/about-aws/global-infrastructure/>

Regions

Availability Zone

Availability Zone

Availability Zone

**Data
Center**

**Data
Center**

**Data
Center**

**Data
Center**

**Data
Center**

**Data
Center**

Edge Location = Data centers placed close to users to deliver content with low latency.

Point of Presence = It's any physical location where a network provider connects users to its network.

AWS Free Tier overview and account safety

Credit-Based Model (New Users)

\$100 free credits + up to \$100 bonus
Valid for 6 months or until used
Upgrade required for full access

Free Trials (Service-based)

Amazon Redshift → ~2 months
Amazon SageMaker → limited trial
Amazon Inspector → ~15 days

Always Free Services (No expiry)

AWS Lambda → 1M requests/month
Amazon DynamoDB → 25 GB storage
Amazon CloudWatch → basic metrics
Amazon SNS → 1M messages

Practicals